



Heuristic Evaluation

of the USDA ACIR Portal

Apex Five

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What is ACIR?

The ACIR portal is a USDA database used by importers, customs brokers, and government officials to look up import requirements for agricultural goods

- Supports over 100,000 country-commodity combinations
- Previous research phases (interviews, survey, comparative evaluation) found issues with search, regulatory language, and information organization

USDA Animal and Plant Health Inspection Services
U.S. DEPARTMENT OF AGRICULTURE

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Agricultural Commodity Import Requirements

APHIS Import Requirements Search Tool for Plant, and Agriculturally Related Products



Plants for Planting and Propagation

Imported material of plants and vegetative parts that are for or capable of propagation, including buds, corms, cuttings, layers, pollen, scions, seeds, tissue, tubers, and like structures



Plants and Plant Products Not for Propagation

Fruits, vegetables, seeds, cut flowers, or handicrafts not for propagation, including products that are fresh, cooked, dried, cut, frozen, and other similar processed states



Wood Products and Byproducts

Logs, lumber, cross ties, firewood, wood chips, mulch, wood packaging material, cork and other similar articles



Pest Organism Permitting

Fungi, insects (including butterflies, bees, and moths), regulated organisms used in pet trade (or smuggled), arthropods (earthworms), parasitic plants, bacteria, noxious weeds and more. See also nonregulated organisms.



Other Items

Tools, materials, or machinery that can act as carriers for pests, including vehicles, containers, bags, packing materials, and other similar articles



Soil, Soil Constituents and Rocks

Compost, manure, rock, sand, soil, and other similar articles



Treatments

Phytosanitary treatments for quarantine-significant plant pests on imported commodities

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USDA.gov

USDA APHIS

Accessibility Statement

USDA Privacy Policy

Non-Discrimination Statement

Information Quality

Whitehouse.gov

FOIA

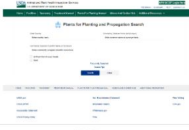
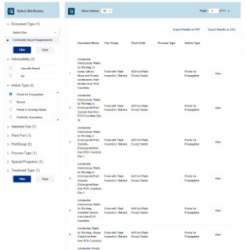
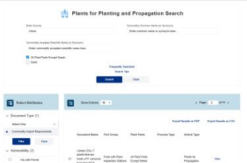
Plain Writing

USA.gov

Our Methods

- Selected 6 of Nielsen's 10 usability heuristics based on relevance to the ACIR portal:
 Visibility of System Status
 Consistency and Standards
 User Control and Freedom
 Recognition Rather than Recall
 Flexibility and Efficiency of Use
 Help and Documentation
- All 5 team members independently evaluated the portal
- Focused on the core task of searching for and interpreting import requirements
- Issues rated on a 0-to-4 severity scale (cosmetic to catastrophic)
- Group debrief to merge duplicates and assign consensus severity ratings

Heuristic Analysis

Link to product website: https://acir.aghis.usda.gov/								Login credentials: N/A		Heuristics to be used: Visibility of System Status, Consistency and Standards, User Control and Freedom, Recognition Rather than Recall, Flexibility and Efficiency of Use, Help and Documentation		Reviewer Names: Shenghao Mei Seorin Jang Dongsub Kim Micaela Ciambone Laylah Campbell	
#	Screen	Problem Identified	Heuristic Used	Severity	Mentioned By	Notes	Screenshots						
1	Homepage / All Pages	Pages go blank for several seconds during transitions with no loading spinner, skeleton screen, or progress message. Users cannot tell if the system is working or has crashed.	1. Visibility	4. Major Issue	4 Reviewers	Dongsub noted blank screens during commodity hierarchy navigation. Laylah flagged no skeleton UI on homepage load. Seorin noted spinning circles with no status info. Micaela noted users lose track of where they are between results and detail pages.	Homepage Page load 						
2	Any Search Title Page	Search requires users to know and enter both a country and the correct commodity name (scientific, common, or synonym). If the input does not match the expected format, the search fails with minimal guidance. Users cannot search by country alone.	6. Recognition	4. Major Issue	5 Reviewers	Flagged by all five evaluators. Seorin noted users must know exactly what to type. Micaela noted the memory burden of needing correct field combinations. Shenghao noted interviewees wanted country-only search.	Search Page 						
3	Any Search Title Page	The autocomplete dropdown disappears when users type a full commodity name, forcing them to backspace until the dropdown reappears.	3. User Control	4. Major Issue	2 Reviewers	Shenghao flagged this from interview findings. Seorin noted the system gives unhelpful error messages when a search fails.							
4	Search Results Page	Filter selections disappear from view as the results list grows. There is no result count, no indicator of active filters, and no visual grouping of results by commodity type or family. Users must scan a long flat list.	6. Recognition	4. Major Issue	5 Reviewers	Dongsub noted filter selections vanish as you scroll. Shenghao noted no active filter indicators. Laylah flagged the flat list with no grouping. Seorin noted users must re-verify items through a commodity list. Micaela noted difficulty tracking position in the decision process.							
5	Search Results Page	There is no way to save a search, bookmark filter combinations, compare two results side by side, or modify one search variable without starting over. Every session requires re-entering all criteria from scratch.	7. Flexibility	4. Major Issue	5 Reviewers	Dongsub described very high interaction cost for comparing items. Seorin noted no customization or saved searches. Laylah flagged no saved filter bookmarks. Micaela and Shenghao both noted limited ability to edit or back out of a search.							
6	Search Results Page	Filters are not shown upfront before the initial search. Users must complete a search first and then discover filtering options afterward.	7. Flexibility	3. Minor Issue	2 Reviewers	Shenghao noted the client proposal specifically asked whether showing filters upfront would improve efficiency. Seorin noted filters require users to open each category manually.							

5 Key Findings

Search

Severity 4, 5/5 evaluators:

Search requires exact terminology. The autocomplete drops off when users type a full name. No country-only search. No way to save or modify a search without starting over.

Help

Severity 4, 5/5 evaluators:

Help content is scattered across multiple pages. Users have to leave their task to find support. Technical terms in documents go unexplained.

Navigation

Severity 3, 5/5 evaluators:

Users lose their place easily. Links behave inconsistently (some open new tabs, some don't). No breadcrumbs. The browser back button breaks search state.

Navigation

Severity 3, 4/5 evaluators:

No loading feedback. Pages go blank during transitions.

Jargon

Severity 3, 4/5 evaluators:

Regulatory jargon on tiles is confusing. Conflicting classifications between tiles led to incorrect guidance in one documented case.

Search
rigidity



Plants for Planting and Propagation Search

Enter Country

Enter country here...

Commodity Common Name (or Synonym)

Enter common name or synonym here...

Commodity Accepted Scientific Name (or Synonym)

Enter commodity accepted scientific name here...

☐ All Plant Parts Except Seeds

☐ Seed

[Frequently Searched](#)

[Search Tips](#)

Search

Clear

No
breadcrumbs



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Commodity Import Requirements

Published Date 5/20/2025, 08:22 AM EDT

Small Lots of Seed, Certain CITES II Orchids, from All Countries Doc 14

[Save PDF](#)

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This document contains current import requirements for the selected article(s) from a designated country or region into one or more designated ports.

+ Header

— Important Information

- These Orchidaceae plants are subject to CITES II, #4, P3. Seeds of these plants are therefore exempt from CITES regulations. See Additional Requirements section below.
- This document relates to Small Lots of Seeds.

Scattered help
resources

Recommendations

Search

Implement faceted search that allows single-variable adjustments and country-only queries. Add persistent autocomplete and a "Save Search" feature.

Help

Consolidate all resources into a single Help Hub. Add hover-over tooltips for technical terms directly within the workflow.

Navigation

Add breadcrumbs and internal navigation buttons. Standardize link behavior so users know what will open in a new tab.

Feedback

Replace blank screens with skeleton loaders or progress indicators.

Jargon

Add plain-language sub-labels under regulatory tiles. Audit the database for conflicting commodity classifications.



Heuristic Evaluation of the USDA ACIR Portal

60+ issues identified, narrowed to 5 key findings & recommendations:

- | | | |
|--------------------------------|---|---|
| 1. Search rigidity | → | Search with persistent autocomplete and save features |
| 2. Scattered help content | → | Single Help Hub with inline tooltips |
| 3. Users losing their place | → | Breadcrumbs and standardized link behavior |
| 4. No loading feedback | → | Skeleton loaders and progress indicators |
| 5. Confusing regulatory jargon | → | Plain-language sub-labels and classification audit |

Usability testing has been completed and we are now analyzing findings and writing the report.

THANK YOU

Introduction:

Purpose:

Evaluate how effectively users navigate, interpret, and apply regulatory information in the Agricultural Commodity Import Requirements (ACIR) portal.

Project Focus:

- Assess navigation, search, and filter usability
- Evaluate clarity and organization of help resources
- Examine mobile experience and small-form-factor use cases

Key Research Areas:

- Efficiency of upfront vs. post-search filtering
- User ability to locate commodity-specific requirements
- Effectiveness of current help pathways (labels, videos, guides)
- Opportunities for a centralized “single help hub”



ACIR Search Tools - Locator Table and Frequently Searched

Are you looking for a way to browse topics in ACIR? Would you like a prebuilt search list of Commodity Import Requirements (CIRs) on a specific topic?

There are two Search Tools available in ACIR:

- **Locator Table** - Found on the [ACIR Help Page](#)
 - Allows users to browse various commodities and locate specialty items in ACIR.
 - Topics are listed alphabetically and include Bags/Bagging, Bee Products, Corn, Fresh Fruits and Vegetables, Herbarium Specimens, Lichen, Live Plants, Plant Oils, Seed Kits, Wooden Handicrafts and much more.
- **Frequently Searched** - Found on all search tile pages, above the Search button
 - Allows users to choose prebuilt searches for quick and easy results based on the tile content.
 - [Plants and Plant Products Not for Propagation](#) has the most topics available (e.g., Seeds, Nuts, and Grains - Fresh Fruit and Vegetables from a certain country - Handicrafts - Mushrooms - Tea Products, etc.)
 - Click on the Select button to the right of the topic to view the search results with commodity import requirements related to the topic.

APEX FIVE



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